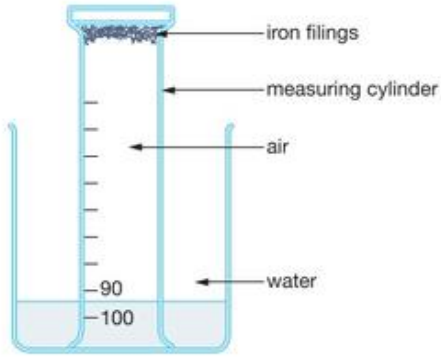
CHAPTER QUESTIONS

SKILLS CRITICAL THINKING 5

1 State the approximate percentage of each of the following gases in dry air:
a nitrogen b oxygen c carbon dioxide d argon

SKILLS PROBLEM SOLVING 7

2 A student carried out an experiment to measure the percentage oxygen in the air by using the rusting of iron. The experimental set-up is shown below. The initial reading on the measuring cylinder was 95 cm^3. They left the apparatus for 2 days and the new reading on the measuring cylinder was 80 cm^3.



a Calculate the percentage oxygen in the air using this data.

SKILLS REASONING, DECISION MAKING

b Explain why the answer is not the same as you had expected and how the experiment could be improved.

SKILLS DECISION MAKING 6

3 A student carries out an experiment to investigate how much gas is produced when various metal carbonates are heated. Their data are shown in the table.

Metal carbonate	Mass of carbonate used/g	Volume of gas collected at rtp/cm^3	Mass of solid left at the end/g
lead(II) carbonate	1.00	89	0.84
copper(II) carbonate	1.00	194	0.64
sodium carbonate	1.00	0	1.00

a Describe an experiment that would allow the student to collect these data.

SKILLS CRITICAL THINKING 5

b Name the gas collected.

SKILLS PROBLEM SOLVING 9

c Write an equation for the reaction occurring with lead(II) carbonate.

SKILLS CRITICAL THINKING 5

d Name the type of reaction occurring with lead(II) carbonate and copper(II) carbonate.

SKILLS REASONING 6

e Explain what these results indicate about the thermal stability of sodium carbonate compared to the other two substances.

7

f Explain why much more gas is given off in the copper(II) carbonate experiment than in the lead(II) carbonate experiment.

SKILLS PROBLEM SOLVING

g Use the data from the lead(II) carbonate experiment to calculate the mass of 100 cm^3 of the gas.

SKILLS EXECUTIVE FUNCTION

h The student repeated the experiment for copper(II) carbonate and only obtained 152 cm^3 of gas this time. Explain two things that could have gone wrong with the second experiment to give this result.

SKILLS CRITICAL THINKING 6

4 Carbon dioxide is described as a *greenhouse gas*. Explain what this means.